

Dynamic routing-scheduling problem for home health care considering caregiver-patient compatibility

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ARTICLE INFO

Keywords:

Routing
Scheduling
Home health care
Unexpected events
Caregiver-patient compatibility
Matheuristics

ABSTRACT

This paper presents a two-stage mathematical formulation for the Home Health Care Routing- Scheduling Problem (HHCRSP) over a planning horizon of multiple days. To devise valid planning, skill requirements and the occurrence of unexpected events are taken into consideration. Caregiver-patient compatibility is another pivotal aspect of the problem, which the present study addresses for the first time. This concept refers to the desirability of the personal relationship between a caregiver and a patient, which plays a key role in fulfilling their satisfaction. The first stage minimizes the Total Planning Distance (TPD), while the second stage maximizes the total compatibility and deals with unexpected events. A health center located in Tehran is considered as a case study to assess the validity of the suggested model and analyze the sensitivity of the objectives with the structural parameters, which in turn leads to some practical managerial insights. Due to the NP-hardness of the studied problem, a matheuristic-based algorithm is applied to reduce the computational time. The computational results approve the proposed solution algorithm's decent performance in obtaining near-optimal solutions quickly.

1. Introduction

Controlling the increase in healthcare expenditures has provoked frequent discussions over recent years. The efficient delivery of healthcare services has imposed high pressure on governmental institutions and public healthcare systems (Shahnejat-Bushehri et al., 2021). For example, the total healthcare costs have surpassed the growth of Gross Domestic Product (GDP) in Canada and its population during the last 13 years (Fathollahi-Fard et al., 2022). In addition, the US healthcare system's cost containment has turned into a challenging concern as the interacting nature of the system is noticeably complex (Xie and Wang, 2017).

As a crucial sector of many healthcare organizations over the globe, Home Health Care (HHC) makes a key contribution to helping patients, particularly the elderly, who should recover after a hospital or need extra support to stay safely at home without unnecessary hospitalization. While patients can be provided with a wide range of healthcare services at the convenience of their homes, the transportation cost and traveling distance between patients' homes need to decrease so that the efficiency of the HHC services is further improved (Fathollahi-Fard et al., 2018; Ziya-Gurabi et al., 2022).

Generally, an HHC system consists of caregivers, patients, and their requests for medical services. The planning of such systems includes assigning the services to the caregivers, which creates the travelling routes for each of them. Therefore, the Vehicle Routing Problem (VRP) and Nurse Rostering Problem (NRP), both of which are the well-known Operations Research problems, form the cornerstones of a Home Health Care Routing-Scheduling Problem (HHCRSP). Usually, patients' requests have starting and ending times, and the caregivers must arrive before the starting time. Thus, the VRP aspect of an HHCRSP can be seen as a VRP with time windows (VRPTW).

The present study is concerned with an HHCRSP in a dynamic environment to reduce the distance traveled by the caregivers. In this setting, the health center's main task is to collect visiting requests from patients and schedule the caregivers to cover all of these requests at patients' homes. This problem's planning is developed within scope of multiple periods (e.g., a week or a month), while considering the number of working days and days off for the caregivers, the maximum and minimum numbers of daily working hours for each of them, and their skill requirements.

Besides devising home visit schedules with short traveling distances, the health center must respond to various unexpected events, such as

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<https://doi.org/10.1016/j.cor.2022.106000>

Received 17 August 2021; Received in revised form 11 April 2022; Accepted 16 August 2022

Available online 20 August 2022

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Table 1
Assumptions and constraints considered in the reviewed studies.

	Objective										NO. of periods						Solution method		
Reference	Dynamic	Rejection	Preference	Continuity of care	Workload balance	No. of caregivers	Overtime	Waiting time	Traveling time/ cost/ distance	Changes in the initial planning	Time windows	Single-period	Multi-period	Compatibility	Skill requirements	Case study country	Exact	(Meta) heuristic	Matheuristic
		Min	Max	Max	Max	Min	Min	Min	Min	Min									
Bennett and Erera (2011)	✓								✓		✓		✓					✓	
Braekers et al. (2016)			✓				✓		✓		✓	✓			✓			✓	
Cappanera and Scutellà (2015)				✓	✓			✓			✓		✓		✓	IT	✓		
Cheng and Rich (1998)									✓		✓	✓						✓	
Cinar et al. (2021)									✓		✓		✓			TR		✓	
Decerle et al. (2019)					✓				✓		✓	✓			✓			✓	
Demirbilek et al. (2019)	✓	✓									✓	✓			✓			✓	
Du et al. (2019)	✓								✓		✓		✓			CN		✓	
Duque et al. (2015)			✓	✓					✓		✓		✓		✓	BE		✓	
Fathollahi-Fard et al. (2019)									✓		✓		✓					✓	
Fikar and Hirsch (2015)					✓				✓		✓	✓			✓	AT			✓
Fikar et al. (2016)	✓							✓	✓		✓	✓						✓	
Grenouilleau et al. (2019)		✓	✓	✓			✓				✓		✓		✓	CA		✓	
Heching et al. (2019)		✓		✓							✓		✓		✓		✓		
Koeleman et al. (2012)	✓	✓							✓				✓					✓	
Lessel (2007)									✓			✓			✓		✓		
Liu et al. (2013)									✓		✓	✓						✓	
Moz and Pato (2004)	✓	✓								✓			✓				✓		
Nickel et al. (2012)		✓		✓			✓		✓		✓		✓		✓	DE		✓	
Nikzad et al. (2021)				✓		✓			✓		✓		✓		✓				✓
Rasmussen et al. (2012)		✓	✓						✓		✓	✓					✓		
Trautsamwieser et al. (2011)			✓				✓	✓	✓		✓		✓		✓	AT		✓	
Thomsen (2006)						✓			✓		✓	✓						✓	
Zhan et al. (2021)								✓	✓			✓						✓	
This study	✓								✓	✓	✓		✓	✓	✓	IR			✓

Note: * AT: Austria; BE: Belgium; CA: Canada; CN: China; DE: Germany; IR: Iran; IT: Italy; TR: Turkey.

newly-added requests, visit cancellations, changes in pre-determined appointment times, or caregivers' availability, to ensure service quality. The unexpected events are exclusive to the second part of the planning horizon, which excludes the first few days. As a result, it is necessary to update the initial planning to be feasible for the second part.

The desirability of the personal relationship between a caregiver and patient is identified as another prominent dimension of service quality (Laferriere, 1993). This factor cannot be estimated unless the caregiver and patient meet in person. Therefore, this paper uses caregiver-patient compatibility data collected over the first few planning days to update the initial planning and satisfy both caregivers and customers.

In the modeling phase of the research, both aspects of the problem (routing and rostering) are simultaneously formulated by the suggested model using a Mixed-Integer Programming (MIP) approach.

Employing an exact algorithm may not be adequate to solve such a complicated mathematical model. As a result, a decomposition-based matheuristic algorithm is adapted to lower the computational time.

The rest of the paper is structured as follows. The related literature is reviewed in Section 2, which culminates into the list of the main novelties of this study. Section 3 defines the problem and the related characteristics in detail. Section 4 explains the solution method. Section 5 carries out a case study and verifies the proposed algorithm's efficiency through a computational experiment. Section 6 extracts some managerial insights from the sensitivity analysis conducted on the case study. In the end, Section 7 provides conclusions and suggestions for future works.

2. Literature review

As mentioned earlier, the HHCRSP involves VRPTW and NRP. The former problem allocates nurses to different vehicle routes respecting patient time windows, while the latter plans duty rosters for each nurse (Lin et al., 2018). A detailed review of an HHCRSP can be found in Cissé et al. (2017) and Fikar and Hirsch (2017).

The majority of works on an HHCRSP address the VRPTW. In this regard, Thomsen (2006) proposed a hybrid Tabu Search (TS)-insertion heuristic method to solve the VRPTW. Rasmussen et al. (2012) adopted branch-and-price for a set partition problem regarded as a VRPTW. Liu et al. (2013) proposed a solution method based on a Genetic Algorithm (GA) and TS for a VRP with simultaneous pickup and delivery. Fikar and Hirsch (2015) studied an HHCRSP and suggested a two-step solution algorithm. The first step of the algorithm finds possible walking routes, and the second step determines the best routing using TS. Fathollahi-Fard et al. (2019) employed five modified versions of Simulated Annealing (SA) to address a bi-objective location-allocation-routing model for Green Home Health Care Supply Chain (GHHCS). Heching et al. (2019) applied an exact optimization method for healthcare service delivery that relies on Logic-based Benders Decomposition (LBBD).

The simultaneous consideration of VRPTW and NRP in the HHCRSP can be seen in several other works. In this regard, Cheng and Rich (1998) studied a VRPTW with part-time and full-time staff in charge of fulfilling patients' requests within fixed time windows associated with each patient. The solution method involves a greedy heuristic that creates staff tours and eventually forms the rosters. Lessel (2007) formed different visit clusters regarding various criteria, such as geographical locations and nurse skills. He proposed a two-step algorithm that dispatches nurses to the clusters in the beginning and then solves a VRPTW for each. Nickel et al. (2012) divided the HHCRSP into NRP, VRP, and nurse re-rostering problem and adopted hybrid Constraint Programming (CP)-Adaptive Large Neighborhood Search (ALNS), CP, and hybrid TS-insertion heuristic to solve each of the problems, respectively. Cappanera and Scutellà (2015) applied Linear Programming (LP) to solve NRP and VRP over a particular planning horizon while balancing nurses' workload and decreasing waiting times between consecutive visits. Duque et al. (2015) employed random Local Search (LS) for the HHCRSP

and took advantage of a two-step algorithm. This algorithm initially maximizes the service level and optimizes the total traveling distance in the end. Decerle et al. (2019) focused on working time balancing in the HHCRSP. They suggested an original hybrid algorithm combining a Memetic Algorithm (MA) and Ant Colony Optimization (ACO) as an efficient algorithm for solving the problem. Also, see Zhan et al. (2021) and Cinar et al. (2021).

When it comes to works that consider dynamic environments caused by unexpected events, in the study of Moz and Pato (2004), the nurse roster of the coming week is released periodically. Some nurses cannot carry out any of the assigned working shifts, new medical services might be requested but are not done by any nurse, some patients decide to cancel their appointments, which leaves the recruited nurses idle. Hence, the nurse roster must be adjusted. It is crucial to bridge the gap between the original and adjusted rosters as the latter would affect the regular schedule of many nurses. Bennett and Erera (2011) developed a rolling horizon myopic planning method for a problem, in which a set of patients revealed dynamically need to be visited based on a prescribed weekly frequency for a particular number of weeks. These weekly visits have to repeat at the same times and days within a patient's service duration. Koeleman et al. (2012) used the Poisson process to simulate the dynamic behavior of patients. They modeled the system as a Markov decision process that aims to minimize the number of patients in the system, and thus, the total rejection and holding costs.

Fikar et al. (2016) proposed a flexible discrete-event-driven meta-heuristic method to deal with dynamic routing-scheduling scenarios in HHC using combined trip sharing and walking policies. In Demirbilek et al. (2019), patients arrive dynamically over time, and acceptance and appointment time decisions must be made immediately. They adopted a novel heuristic based on defining different scenarios, including actual and randomly generated demands, responding to new clients with a straightforward but quick heuristic, and analyzing results to determine whether to accept the new patient and at which appointment day/time. Du et al. (2019) assumed visit cancellation, call for emergency aid, and medical tool breakdown as the unexpected incidents that cause scheduling conflicts. They established an effective real-time scheduling model to minimize the total time needed for scheduling and proposed an improved variant of the MA to optimize the model. Also, see Srinivasan and Sangeetha (2020).

Regarding studies focused on caregivers' or patients' satisfaction, the former aspect is about penalizing assignments to patients whom the associated caregiver does not wish to visit. The latter aspect involves penalizing deviations from preferred visit times or caregivers. Traut-samwieser et al. (2011) employed several terms in the objective function to minimize nurses' and clients' dissatisfaction. These terms include overtime working, unfulfilled preferences of clients, soft time window violations, recruitment of overqualified nurses, and unpaid driving times.

Braekers et al. (2016) analyzed a trade-off between cost and patient convenience, calculated by the deviation from the preferred visit time and how disfavored the assigned nurses are. According to the results, starting with the lowest cost solution, the average service level might improve noticeably by spending small extra charges. Grenouilleau et al. (2019) introduced specialty and language as patients' mandatory and optional requirements, respectively. They also included caregivers' characteristics (e.g., gender) in their skills to be respected while assigning caregivers to the visits. The study of Nikzad et al. (2021) shows that districting/zoning sets of patients and caregivers before making assignment, scheduling, and routing decisions, helps caregivers to deliver care to the patients in their cluster more quickly, which ultimately increases the patients and the caregivers' satisfaction.

Table 1 shows the assumptions and constraints considered in the related works, based on which the following information can be extracted:

Table 2

Requested services over a specific day.

Patient	Service 1	Service 2	Service 3
J_1	8:00-9:00	14:00-15:00	-
J_2		No request	
J_3	9:00-11:00	-	-
J_4	8:00-9:00	19:00-20:00	-
J_5		No request	
J_6	7:00-8:00	20:00-21:00	-
J_7	12:00-13:00	19:00-20:00	-
J_8	10:00-11:00	15:00-16:00	20:00-21:00
J_9	16:00-18:00	-	-
J_{10}		No request	

- Minimizing the traveling time/cost/distance and the number of rejected requests are two of the most prevalent objective functions in the HHCRSP, whereas maximizing workload balance, minimizing the number of caregivers, and bridging the gap between the planings in dynamic environments are not sufficiently studied. Other relatively common objectives are maximizing continuity of care and minimizing staff overtime and patient waiting time.
- Although patient preference -in terms of caregivers' personal characteristics, such as gender, language, certificates, etc.- is respected in several studies, none of the papers has shed light on the abstract concept of caregiver-patient compatibility, which unlike preference, is not easily estimable.
- Time windows and the skill requirements of the caregivers are important constraints considered by almost every HHRSP study.
- There seems to be a balanced focus on single-period and multi-period problems, while dynamism is mainly considered in the latter type due to the negligible probability for the occurrence of unexpected events over short time intervals.
- Despite the activity of several HHC centers in Iran and the rise in the demand for HHC services by the growing population of senior citizens, none of the reviewed papers has studied the HHCRSP in this country.
- (Meta)heuristic and exact solution methods are frequently used in the HHCRSP; however, as semi-exact methods, matheuristic approaches are rarely employed.

According to the explanations above, the main contributions of this paper are as follows:

1. Introducing the notion of compatibility as an influential factor in achieving caregiver's and patients' satisfaction.
2. Developing a more comprehensive two-stage mathematical model that incorporates the occurrence of unexpected events, skill requirements, working regulations, and patient time windows in a multi-period HHCRSP.
3. Adapting a matheuristic-based algorithm to cope with the complexity of large-sized problem instances.
4. Conducting a case study in a health center located in Tehran, Iran, to check out the model's performance, analyze the influence of the structural parameters on the objectives, and infer some informative managerial insights in the end.

3. Problem statement

The planning of an HHC center caregiver is considered to be optimized by applying mathematical programming. Caregivers consist of nurses and auxiliary nurses. As nurses are more qualified than auxiliary nurses, they can conduct a wider range of services. Table 2 shows a sample of data collected by the center reporting services requested by patients. Each service is indicated by its starting and ending times, determining the service duration. These services need to be conducted by the available caregivers using the optimum plan.

Table 3

Structured services requested by patients over a specific day.

Day	Period	Patient									
		J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8	J_9	J_{10}
D_1	P_1	0	0	0	0	0	1	0	0	0	0
	P_2	1	0	0	1	0	0	0	0	0	0
	P_3	0	0	1	0	0	0	0	0	0	0
	P_4	0	0	1	0	0	0	0	1	0	0
	P_5	0	0	0	0	0	0	0	0	0	0
	P_6	0	0	0	0	0	0	1	0	0	0
	P_7	0	0	0	0	0	0	0	0	0	0
	P_8	1	0	0	0	0	0	0	0	0	0
	P_9	0	0	0	0	0	0	0	1	0	0
	P_{10}	0	0	0	0	0	0	0	0	1	0
	P_{11}	0	0	0	0	0	0	0	0	1	0
	P_{12}	0	0	0	0	0	0	0	0	0	0
	P_{13}	0	0	0	1	0	0	1	0	0	0
	P_{14}	0	0	0	0	0	0	0	1	0	0

Over a working day, caregivers start their travel from the home of the first patient assigned to each of them, and finally finish their travel at the last patient's home. Each service lasts between one and two hours, and caregivers work six to eight hours per day. In the case of serving a patient in a period, the caregiver cannot accept a new visit in the next period since a one-period traveling time is assumed for commuting between the patients' homes. Another assumption is that when a caregiver is assigned more than one task during a day, he/she waits at a location near the home of the current patient until it is the time to leave for the next service. According to Table 2, some patients do not request any services, while others ask for one, two, or three services during a given day.

This problem, considered as a sequence of daily schedules, is formulated using an MIP model. As a result, all parameters and variables must be integer or binary. However, as Table 2 demonstrates, the parameters of the requested services are neither integer nor binary. To apply this data in the MIP model, it is converted into numerical parameters demonstrated by Table 3. Other parameters, such as the planning duration and the number of working hours, are integers. This table is an instance of the *theRS* matrix that includes the parameters of the requested services in which P_1 and P_{14} denote the first (7:00-8:00) and last (20:00-21:00) period of a working day, respectively.

It is worth mentioning that firstly, the data in Table 2 is based on the real-world case study in this research. As this table shows, the studied health center assigns a single/multiple one-hour periods to the requested services, which provides the opportunity to break each working day into 14 one-hour periods and feed the requests data into an MIP model accordingly to solve the HHCRSP for the center. Secondly, in case of having a system where the starting/ending times or service durations are not limited to integers for hours (e.g., patient 1's starting time and ending times are 8:15 and 9:30), it is possible to break the days into shorter periods (e.g., quarter-hour periods or five-minute periods) and use the same mathematical formulations.

The occurrence of unexpected events (dynamism), such as newly-added requests, visit cancellations, changes in pre-determined appointment times, or caregivers' availability, causes a significant decrease in the initial planning's performance. Consequently, updated planning is needed to cope with such changes.

For example, consider a health center that provides the caregivers with an initial working plan for the original planning horizon at the beginning of each horizon. Dynamism is very low for short horizons, and it increases exponentially as the planning duration extends. Therefore, it is assumed that the initial planning is applicable for the first few days (i.e., the first part) of the planning horizon, while the updated planning addresses the changes over the rest of the days (i.e., the second part).

During the first part of the horizon, the center has the opportunity to track the second part's unexpected events, which means it can register new requests and document cancellations or changes in appointment

times for the coming days. It can also update the caregiver availability during the second part of the planning horizon.

Implementing the initial planning for a fraction of the planning days makes dealing with unexpected events possible and provides the HHC center with a precious opportunity to collect compatibility data for caregiver-patient pairs who meet each other over this time.

At the end of each service, caregivers and patients are asked to rate each other from 0 (i.e., very low satisfaction) to 10 (i.e., very high satisfaction) through their accounts on the health center's website. This rating is based on the quality of care, communication, and relationships. The average rating between each pair of caregiver-patient results in the caregiver-patient compatibility data that is another basis to update the initial planning. The mediocre compatibility level of five is presumed for other pairs who do not have the chance to meet in person during the first part of the planning horizon.

It should be noted that the initial and the updated planning's mathematical models are almost the same; however, they have different planning horizons and objectives. The initial planning covers the original planning horizon, and its objective is to minimize the total traveling distance. On the other hand, the updated planning excludes the first few days since it is devised during the studied horizon. Moreover, it aims to maximize the total caregiver-patient compatibility to increase their satisfaction while minimizing the deviation of the updated planning from the initial one so that the new planning is as accomplishable as possible for the caregivers.

3.1. Problem formulation

A two-stage MIP model is proposed to formulate the above-mentioned problem. The first stage determines the initial planning, while the second stage is in charge of updating it according to the unexpected events and caregiver-patient compatibility data. The employed indices, parameters, and variables are presented below. In the succeeding sections, the prime symbol applied on the top of some parameters signifies that the updated version of the parameter needs to be used.

Sets:	
$I = \{1, 2, \dots, n\}$	Caregivers, $i, l \in I$.
$J = \{1, 2, \dots, m\}$	Patients, $j, k \in J$.
$P = \{1, 2, \dots, h\}$	Periods, $p \in P, P^- = \{0, 1, \dots, h-1\}$.
$D = \{1, 2, \dots, t\}$	Planning days, $d \in D, \hat{D} = \{e+1, \dots, t\}$, where $\{1, \dots, e\}$ is the first part of the horizon.
$S = \{1, 2, \dots, r\}$	Services, $s \in S, S^- = \{0, 1, \dots, r-1\}$.
Parameters:	
MP	Maximum number of permitted working periods during a day.
DI_{jk}	Distance between patient j and patient k .
η_i	Binary; 1 if caregiver i is a nurse and 0 if the caregiver is an auxiliary nurse.
ρ_{jpd}	Binary; 1 if a nurse is required to visit patient j in period p of day d and 0 if an auxiliary nurse can handle the service.
RS_{jpd}	Binary; 1 if patient j asks for service in period p of day d ; and 0, otherwise.
CN_{ij}	Rating given by caregiver i to patient j .
CP_{ij}	Rating given by patient j to caregiver i .
CNP_{ij}	Compatibility of caregiver i and patient j .
PE_i	Penalty of change in caregiver i 's working plan compared to the initial planning.
OFF_{id}	Binary; 1 if caregiver i takes day d off; and 0, otherwise.
M	A very large positive number.
Decision variables:	
x_{ijpd}	Binary; 1 if caregiver i visits patient j in period p of day d ; and 0, otherwise.
y_{ijpd}	Binary; 1 if period p of day d is the first period of a service asked by patient j and it is done by caregiver i .
u_{ijpd}	Integer; if y_{ijpd} equals 1, it represents the ordinal number of services carried out by caregiver i on day d .
v_{ijds}	Binary; 1 if s th service of caregiver i on day d is for patient j ; and 0, otherwise.
γ_{ijkds}	

(continued on next column)

(continued)

	Binary; 1 if caregiver i visits patient j and k as the s th and $(s+1)$ th service, respectively.
TD_{id}	Traveling distances of caregiver i on day d .
TT	Total Planning Distance (TPD).

3.2. First stage: Mathematical formulation of the initial planning

The initial planning is modeled as follows:

$$\text{Min} Z_1 = TT \quad (1)$$

s.t.

$$\sum_{i=1}^n x_{ijpd} = RS_{jpd} \quad \forall j, p, d \quad (2)$$

$$\sum_{j=1}^m x_{ijpd} \leq 1 \quad \forall i, p, d \quad (3)$$

$$\sum_{j=1}^m \sum_{p=1}^h x_{ijpd} \leq MP \quad \forall i, d \quad (4)$$

$$x_{ijpd} + \sum_{k \in J-j} x_{ik(p+1)d} \leq 1 \quad \forall i, j, p \in P^-, d \quad (5)$$

$$x_{ijpd} + \sum_{l \in I-i} x_{lj(p+1)d} \leq 1 \quad \forall i, j, p \in P^-, d \quad (6)$$

$$x_{ijpd} \leq \eta_i - \rho_{jpd} + 1 \quad \forall i, j, p, d \quad (7)$$

$$y_{ijpd} = x_{ijpd} \times (1 - x_{ij(p-1)d}) \quad \forall i, j, p \in \{P | P \geq 2\}, d \quad (8)$$

$$y_{ij1d} = x_{ij1d} \quad \forall i, j, d \quad (9)$$

$$u_{ijpd} = y_{ijpd} \times \sum_{k=1}^m \sum_{q=1}^p y_{ikqd} \quad \forall i, j, p, d \quad (10)$$

$$\sum_{p=1}^h u_{ijpd} = \sum_{s=1}^r s \times v_{ijds} \quad \forall i, j, d \quad (11)$$

$$\sum_{j=1}^m v_{ijds} \leq 1 \quad \forall i, d, s \quad (12)$$

$$\sum_{j=1}^m v_{ijd(s+1)} \leq \sum_{j=1}^m v_{ijds} \quad \forall i, d, s \in S^- \quad (13)$$

$$\gamma_{ijkds} = v_{ijds} \times v_{ikd(s+1)} \quad \forall i, j, k, d, s \quad (14)$$

$$TD_{id} = \sum_{j=1}^m \sum_{k=1}^m \sum_{s=1}^r DI_{jk} \times \gamma_{ijkds} \quad \forall i, d \quad (15)$$

$$TT = \sum_{d=1}^t \sum_{i=1}^n TD_{id} \quad (16)$$

$$x_{ijpd}, y_{ijpd} \in \{0, 1\} \quad \forall i, j, p, d \quad (17)$$

$$u_{ijpd} \in \text{Integer} \quad \forall i, j, p, d \quad (18)$$

$$v_{ijds} \in \{0, 1\} \quad \forall i, j, d, s \quad (19)$$

$$\gamma_{ijkds} \in \{0, 1\} \quad \forall i, j, k, d, s \quad (20)$$

The objective function (1) is in charge of minimizing the TPD. Constraints (2) ensure the assignment of one and only one caregiver to

each service. According to Constraints (3), caregivers cannot handle more than one patient in a period. Constraints (4) prevent caregivers from working more than the permitted daily working hours. Constraints (5) guarantee that in case of serving a patient in a period, the caregiver is not allowed to accept a new visit in the next period since a one-period traveling time is assumed for commuting between the patients' homes. Based on Constraints (6), services that last for more than one period need to be carried out by the same caregiver. Constraints (7) control the dedication of the qualified caregivers to each service. Constraints (8) and (9) change the decision variable x_{ijpd} to y_{ijpd} . As Constraints (8) are non-linear and should be replaced by their linearized equivalents presented in Appendix A.

Constraints (10) presents another shift from the decision variable y_{ijpd} to u_{ijpd} to record the ordinal number of the daily services carried out by a caregiver. For instance, $u_{1473} = 2$ declares that caregiver 1 visits patient 4 in the seventh period of day 3 as his second daily task. It should be noted that the index of the period only signifies the starting period of service. Hence, if the mentioned example is related to a three-period service, then $u_{1473} = 2$ but $u_{1483} = 0$ and $u_{1493} = 0$. The linearized form of these constraints can be found in Appendix A.

Constraints (11) and (12) change the decision variable u_{ijpd} (assignment of caregivers to daily periods) to v_{ijds} (assignment of caregivers to services). For example, $v_{2453} = 1$ signifies that the third service of caregiver 2 on day 5 is for patient 4. Constraints (13) control the ordinal nature of variables v_{ijds} according to index s . In other words, for a caregiver on a working day, the second service cannot be activated when the first service is inactive. Now, by employing variables v_{ijds} , the sequence of visits for each caregiver can be determined.

Variables formulated by Constraints (14) imply travel between two patients for a caregiver, which in turn transfers the assignment problem to the routing problem. This set of non-linear constraints needs to be replaced by their linearized equivalents presented in Appendix A. The daily traveling distance of each caregiver is obtained by Constraints (15). Constraints (16) calculate the TPD, the summation of the caregivers' daily traveling distances during the studied planning horizon. Finally, Constraints (17) to (20) imply the domain of the proposed decision variables.

3.3. Second stage: Mathematical formulation of the updated planning

Although the formulations of the initial and the updated planning are roughly the same, the changes listed below must be applied in the model presented in the previous section so that it applies to the updated planning:

1. Sets J , D , and S have to be replaced with $\hat{J}$, $\hat{D}$, and $\hat{S}$, respectively. This is because the updated planning only covers the second part of the planning horizon. Also, the former sets are no longer useful due to changes in the patients' requests.
2. The objective functions (21) and (22) substitute for objective (1) and lead to a bi-objective mathematical model for the updated planning. The objective function (21) aims at maximizing the total caregiver-patient compatibility by assigning caregivers to patients with whom they get along well. On the other hand, the objective function (22) tries to minimize the deviation of the updated planning since a high deviation from the initial planning prevents the caregivers from following the new plan. To this end, a penalty coefficient PE_i is defined to keep the assignment of caregivers to the services that are not affected by dynamism unchanged if possible.
3. x_{ijpd} for days during the second part of the planning horizon ($d \in \hat{D}$) that results from the initial planning, is used as an input parameter ($x_{ijpd}^{\hat{A}}$) in the updated planning.

4. The parameter RS_{ijpd} is replaced by $\hat{RS}_{ijpd}$ to devise the updated planning based on the revised table of requests.
5. Constraints (24) are added to the model to ensure that caregivers are not assigned to any service on their off days.

$$\text{Max} Z_2 = \sum_{i \in I} \sum_{j \in \hat{J}} \sum_{p \in \hat{P}} \sum_{d \in \hat{D}} CNP_{ij} \times x_{ijpd} \quad (21)$$

$$\text{Min} Z_3 = \sum_{i \in I} \sum_{j \in \hat{J}} \sum_{p \in \hat{P}} \sum_{d \in \hat{D}} PE_i \times |x_{ijpd} - x_{ijpd}^{\hat{A}}| \quad (22)$$

The objective function (22) is non-linear as it consists of an absolute term. Thus, it needs to be written in the form of the objective function (23) accompanied by Constraints (25) and (26).

Regarding the above-mentioned differences between the initial and updated planning, the final formulation of the updated planning as an MIP model is as follows:

Objective (21)

$$\text{Min} Z_3 = \sum_{i \in I} \sum_{j \in \hat{J}} \sum_{p \in \hat{P}} \sum_{d \in \hat{D}} PE_i \times (x_{ijpd}^+ - x_{ijpd}^-) \quad (23)$$

s.t.

Constraints (2) to (7), Constraints (A. 1) to (A. 3), Constraints (9), Constraints (A. 4) to (A. 10), Constraints (11) to (13), Constraints (A. 11) to (A. 13), Constraints (15) to (20).

$$\sum_{j \in \hat{J}} \sum_{p \in \hat{P}} x_{ijpd} \leq M \times (1 - OFF_{id}) \quad \forall i, d \in \hat{D} \quad (24)$$

$$x_{ijpd} - x_{ijpd}^{\hat{A}} = x_{ijpd}^+ - x_{ijpd}^- \quad \forall i, j \in \hat{J}, p, d \in \hat{D} \quad (25)$$

$$x_{ijpd}^+, x_{ijpd}^- \in \{0, 1\} \quad \forall i, j \in \hat{J}, p, d \in \hat{D} \quad (26)$$

4. Solution method

A semi-exact combinatorial solution method is employed to solve the two-stage routing-scheduling model presented in Section 3. Different steps of this algorithm are listed below:

1. Solve the initial planning model using the matheuristic algorithm.
2. Solve the updated planning model using the matheuristic algorithm and the outputs of the initial planning.
3. Use AUGMECON2 to obtain non-dominated Pareto (NDP) solutions for the bi-objective model resulted from Step 2.

The AUGMECON2 method mentioned in the third step is a well-known exact algorithm for solving multi-objective problems (Mavrotas and Florios, 2013). Compared to AUGMECON1, AUGMECON2 uses the lexicographic optimization procedure to produce a pay-off table, which leads to an influential generation of Pareto-optimal solutions (Babaei Tirkolaee et al., 2019; Ghodrattnama et al., 2019). A comprehensive explanation of the adapted matheuristic algorithm is provided in the next section.

4.1. Matheuristic algorithm

The proposed mathematical model incorporates different aspects of routing, scheduling, and assignment problems simultaneously. It aims to determine:

1. The required number of caregivers to conduct all of the services each day and during the entire planning horizon.
2. Services for which each caregiver is responsible.

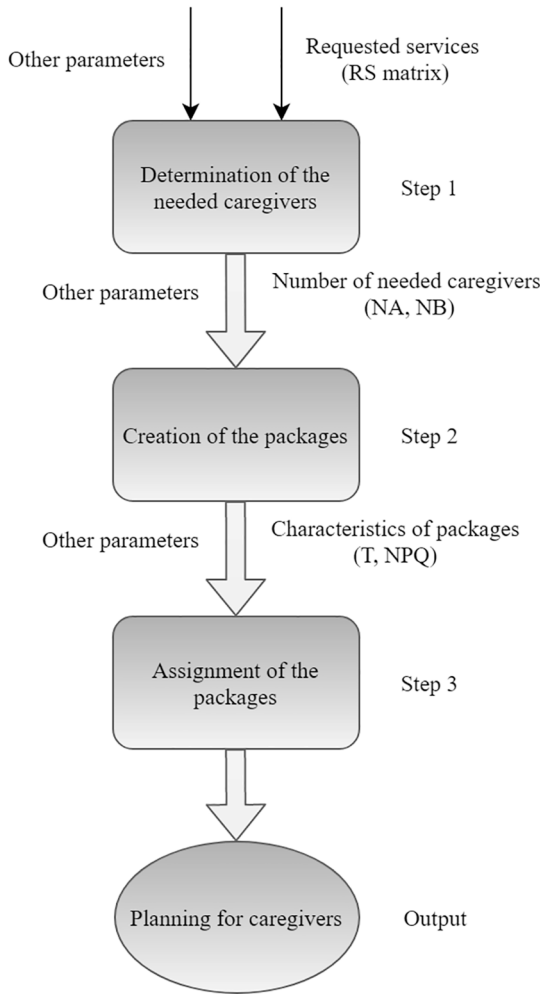


Fig. 1. Matheuristic algorithm for the HHCRSP.

3. Working periods of each caregiver during a day.
4. The routes traveled by each caregiver during the planning days.

The simultaneous calculation of all these four variables significantly increases the complexity of the problem, necessitating several less complicated models resulting from the primary model's decomposition.

This paper applies a modified version of the matheuristic algorithm suggested by Moussavi et al. (2019). Also, see Vahedi-Nouri et al. (2021) and Fatemi-Anaraki et al. (2021). Fig. 1 demonstrates the overall structure of the applied matheuristic algorithm.

The number of the required caregivers (NA nurses and NB auxiliary nurses) to conduct all services is determined in the first step according to the matrix of services (RS). The objective is to minimize the total number of caregivers, used as input data in the second step.

Several service packages are created for each planning day in the second step. These packages are later assigned to the caregivers regarding their qualification levels and the daily working hour restrictions. For example, for the first planning day, one service of patient 1 (8:00-9:00), one service of patient 8 (15:00-16:00), and one service of patient 6 (20:00-21:00) create a package for this day. In other words, the visiting sequence of patients (routes) is determined for caregivers during each day. Consequently, each daily package's traveling distance must be minimized as it is assigned to a caregiver. The traveling distance (T_{gd}) and the number of working hours (NPQ_{gd}) are the main characteristics of each package. These outcomes are the input parameters of the next step.

In the final step, the daily service packages need to be assigned to the caregivers. For instance, suppose the first and third packages of two

consecutive days are assigned to a caregiver. The packages have a traveling distance of 16 km and 20 km, respectively. In this case, his traveling distance over these two days is 36. This step determines which package has to be assigned to which caregiver each day. As the final step of the algorithm, this step shares the same objective with the integrated MIP model. The following sections present the MIP models of the steps mentioned above.

4.2. First step of the matheuristic algorithm for the initial planning

The additional model elements employed to formulate the first step of the matheuristic algorithm are as follows:

Parameters:

RI_{jp} Extraction of matrix RS showing only the first period of each service (See an example in Fig. 2).

Decision variables:

CA_{pd} Integer; a number of nurses required in period p of day d to conduct all services that must be done by a nurse in this period.

CB_{pd} Integer; a number of auxiliary nurses required in period p of day d to conduct all services that must be done by an auxiliary nurse in this period.

EPA_d Integer; a required number of nurses to conduct services that must be done by a nurse on day d regarding parallel daily services.

EPB_d Integer; a required number of auxiliary nurses to conduct services that must be done by an auxiliary nurse on day d regarding parallel daily services.

ETA_d Integer; a required number of nurses to conduct services on day d regarding permitted daily working hours.

ETB_d Integer; a required number of auxiliary nurses to conduct services on day d regarding permitted daily working hours.

NA Integer; a number of nurses required to conduct all services that must be done by a nurse during the planning horizon.

NB Integer; a number of auxiliary nurses required to conduct all services that must be done by an auxiliary nurse during the planning horizon.

NT Integer; a number of caregivers required to conduct all services during the planning horizon.

The MIP model for the first step is formulated by:

$$\text{Min } NT = NA + NB \quad (27)$$

s.t.

$$A_{pd} = \sum_{j \in J} RS_{jpd} \times \rho_{jpd} + \sum_{j \in J} RI_{j(p+1)d} \times \rho_{jpd} \quad \forall p \in P - \{h\}, d \quad (28)$$

$$CA_{pd} = \sum_{j \in J} RS_{jpd} \times \rho_{jpd} \quad \forall p = h, d \quad (29)$$

$$EPA_d = \text{Max}_p \{ CA_{pd} \} \quad \forall d \quad (30)$$

$$ETA_d = \sum_{j \in J} \sum_{p \in P} \frac{RS_{jpd}}{MP} \times \rho_{jpd} \quad \forall d \quad (31)$$

$$NA = \text{Max} \{ \text{Max}_d \{ EPA_d \}, \text{Max}_d \{ ETA_d \} \} \quad (32)$$

$$CB_{pd} = \sum_{j \in J} RS_{jpd} \times (1 - \rho_{jpd}) + \sum_{j \in J} RI_{j(p+1)d} \times (1 - \rho_{jpd}) \quad \forall p \in P - \{h\}, d \quad (33)$$

$$CB_{pd} = \sum_{j \in J} RS_{jpd} \times (1 - \rho_{jpd}) \quad \forall p = h, d \quad (34)$$

$$EPB_d = \text{Max}_p \{ CB_{pd} \} \quad \forall d \quad (35)$$

$$ETB_d = \sum_{j \in J} \sum_{p \in P} \frac{RS_{jpd}}{MP} \times (1 - \rho_{jpd}) \quad \forall d \quad (36)$$

$$NB = \text{Max} \{ \text{Max}_d \{ EPB_d \}, \text{Max}_d \{ ETB_d \} \} \quad (37)$$

$$CA_{pd}, CB_{pd} \in \text{Integer} \quad \forall p, d \quad (38)$$

$$EPA_d, EPB_d, ETA_d, ETB_d \in \text{Integer} \quad \forall d \quad (39)$$

$$NA, NB, NT \in \text{Integer} \quad (40)$$

The objective function (27) aims to minimize the number of caregivers required to conduct all services during the planning horizon, which is the summation of the required nurses and auxiliary nurses over this time. Constraints (28) and (29) calculate the number of periodic patients who have to be visited by a nurse and thus, the required number of nurses in each period. Constraints (30) obtain the number of daily nurses required based on the maximum number of periodic nurses over the day. According to Constraints (31), the number of daily nurses required equals the total hours of daily services that have to be done by nurses, divided by the maximum number of permitted daily working hours for each caregiver. As a result, between EPA_d and ETA_d , the variable with the maximum value determines the real number of the required nurses for the day. Constraints (32) obtain the number of required nurses during the planning horizon using the maximum daily needs. Likewise, the number of required auxiliary nurses during the planning horizon is calculated using Constraints (33) to (37). Constraints (38) to (40) denote the domain of the proposed decision variables.

Now that the required number of nurses and auxiliary nurses are determined, the daily service packages each of which encompasses several services can be created. Services of the same package must be on the same day and require the same qualification level. It should be noted that the working hours of a package must not outnumber the maximum permitted working hours of a caregiver. Moreover, the packages on each day must not outnumber daily caregivers. In the next section, a mathematical formulation is provided to create daily service packages.

4.3. Second step of the matheuristic for the initial planning

The number of nurses and auxiliary nurses obtained in the previous section is the second step's input parameters. Other model elements employed to formulate the second step of the matheuristic are as follows:

Sets:	
$GA = \{1, 2, \dots, NA\}$	Service packages that must be done by a nurse.
$GB = \{1, 2, \dots, NB\}$	Service packages that must be done by an auxiliary nurse.
$G = \{1, 2, \dots, NT\}$	All service packages, $GA \cup GB = G, g \in G$.
$W = \{1, 2, \dots, MS\}$	Services in one package, $w \in W$.
Parameters:	
RIN_{jps}	Extraction of matrix RI showing the starting periods of the services in the order of the number of services for each patient (See an example in Fig. 3).
PS_{jds}	The number of periods that the s th service on patient j during day d lasts.
MS	Maximum number of permitted daily services for a caregiver.
Decision variables:	
B_{gwjds}	Binary; 1 if the s th service of patient j on day d represents the w th service of package g ; and 0; otherwise.
NPQ_{gd}	Integer; number of working periods for package g on day d .
F_{gwjkd}	Binary; 1 if the w th and the $(w+1)$ th service of package g on day d are for patient j and patient k , respectively; and 0, otherwise.
T_{gd}	Traveling distance of package g on day d .
T^{max}	Maximum traveling distance among the packages.

The MIP model for the second step is formulated by:

$$\text{Min } T^{max} = \text{Max}_{gd} \{T_{gd}\} \quad (41)$$

s.t.

$$\sum_{g \in GA} \sum_{w \in W} B_{gwjds} = \sum_{p \in P} RIN_{jps} \times \rho_{jpd} \quad \forall j, d, s \quad (42)$$

$$\sum_{g \in GB} \sum_{w \in W} B_{gwjds} = \sum_{p \in P} RIN_{jps} \times (1 - \rho_{jpd}) \quad \forall j, d, s \quad (43)$$

$$\sum_{j \in J} \sum_{s \in S} B_{gwjds} \leq 1 \quad \forall g \in G, w, d \quad (44)$$

$$\sum_{w \in W} \sum_{j \in J} \sum_{s \in S} B_{gwjds} \geq 1 \quad \forall g \in G, d \quad (45)$$

$$\sum_{w \in W} \sum_{j \in J} \sum_{s \in S} B_{gwjds} \leq MS \quad \forall g \in G, d \quad (46)$$

$$\sum_{w \in W} \sum_{j \in J} \sum_{s \in S} B_{gwjds} \times PS_{jds} \leq MP \quad \forall g \in G, d \quad (47)$$

$$NPQ_{gd} = \sum_{w \in W} \sum_{j \in J} \sum_{s \in S} B_{gwjds} \times PS_{jds} \quad \forall g \in G, d \quad (48)$$

$$\sum_{j \in J} \sum_{s \in S} B_{g1jds} \geq \sum_{j \in J} \sum_{s \in S} B_{g(g+1)jds} \quad \forall g \in G - \{NT\}, d \quad (49)$$

$$\sum_{j \in J} \sum_{s \in S} B_{gwjds} \geq \sum_{j \in J} \sum_{s \in S} B_{g(w+1)jds} \quad \forall g \in G, w \in W - \{MS\}, d \quad (50)$$

$$F_{gwjkd} \leq \sum_{s \in S} B_{gwjds} \quad \forall g \in G, w, j, k, d \quad (51)$$

$$F_{gwjkd} \leq \sum_{s \in S} B_{g(w+1)kds} \quad \forall g \in G, w \in W - \{MS\}, j, k, d \quad (52)$$

$$F_{gwjkd} \geq \sum_{s \in S} B_{gwjds} + \sum_{s \in S} B_{g(w+1)kds} - 1 \quad \forall g \in G, w \in W - \{MS\}, j, k, d \quad (53)$$

$$T_{gd} = \sum_{w \in W} \sum_{j \in J} \sum_{s \in S} F_{gwjkd} \times DI_{jk} \quad \forall g \in G, d \quad (54)$$

$$B_{gwjds} \in \{0, 1\} \quad \forall g \in G, w, j, d, s \quad (55)$$

$$F_{gwjkd} \in \{0, 1\} \quad \forall g \in G, w, j, k, d \quad (56)$$

$$NPQ_{gd} \in \text{Integer} \quad \forall g \in G, d \quad (57)$$

Objective function (41) minimizes the maximum traveling distance of the packages to balance the traveling distance among them. Constraints (42) and (43) ensure that every single service is assigned to a package, and all services of a package require the same qualification level. Based on Constraints (44), each position of a package must be filled by at most one service. Constraints (45) guarantee that no empty daily package is created. Constraints (46) restrict the number of services in each package. According to these constraints, a daily package cannot encompass more than the permitted number of services for a caregiver over a day. Using Constraints (47), the working hours of packages are restricted as well. Constraints (49) control the precedence of packages in terms of service allocation. Regarding these constraints, the second package can only be allocated after the first package; the third package cannot be allocated before the second one, and so on. Likewise, the order of assigning the positions inside a package is controlled by Constraints (50). Constraints (51) to (53) determine movement patterns of the packages and stand for the fact that there is travel between patient j and k after the w th service of package g if and only if the w th and $(w+1)$ th position of this package are filled by a service on patient j and k , respectively. The traveling distances of packages are measured by Constraints (54). Constraints (55) to (57) denote the domain of the proposed decision variables.

In this step, the service packages are created. For each planning day, the number of packages needs to be the same as the number of the required caregivers, which is determined in the first step of the

algorithm. The third step addresses the assignment of daily packages to the caregivers.

4.4. Third step of the matheuristic algorithm for the initial planning

The variables T_{gd} and NPQ_{gd} obtained in the previous section are used as the parameters of the third step. Other model elements employed to formulate the third step of the matheuristic algorithm are as follows:

Sets:	
$IA = \{1, 2, \dots, NA\}$	Nurses, $IA \subset I$.
$IB = \{1, 2, \dots, NB\}$	Auxiliary nurses, $IB \subset I$.
Parameters:	
H	Coefficient to convert the number of periods into the number of hours.
MD	Maximum number of permitted working days for a caregiver over the planning horizon.
UH	Maximum number of permitted working hours for a caregiver during the day.
LH	Minimum number of permitted working hours for a caregiver during a day.
CG_{ig}	Compatibility of caregiver i and package g (sum of CNP_{ij} s, where j is a patient whose requested service is in package g).
Decision variables:	
A_{igd}	Binary; 1 if caregiver i is allocated to package g on day d and 0 otherwise.

The MIP model for the third step is formulated by:

Min Z_1

s.t.

$$\sum_{i \in IA} A_{igd} = 1 \quad \forall g \in GA, d \quad (58)$$

$$\sum_{i \in IB} A_{igd} = 1 \quad \forall g \in GB, d \quad (59)$$

$$\sum_{g \in G} A_{igd} \leq 1 \quad \forall i \in I, d \quad (60)$$

$$\sum_{g \in G} \sum_{d \in D} A_{igd} \leq MD \quad \forall i \in I \quad (61)$$

$$\sum_{g \in G} \sum_{d \in D} A_{igd} \times PQ_{gd} \times H \leq UH \quad \forall i \in I \quad (62)$$

$$\sum_{g \in G} \sum_{d \in D} A_{igd} \times NPQ_{gd} \times H \geq LH \quad \forall i \in I \quad (63)$$

$$TD_{id} = \sum_{g \in G} T_{gd} \times A_{igd} \quad \forall i \in I, d \quad (64)$$

Constraints (16)

$$A_{igd} \in \{0, 1\} \quad \forall i \in I, g \in G, d \quad (65)$$

This step's objective function is the same as the objective function of the integrated MIP model, which in turn makes the comparison between the matheuristic algorithm and the exact algorithm possible. Constraints (58) and (59) ensure that each package is assigned to one and only one caregiver who is qualified to carry out all of the services in the package. According to Constraints (60), caregivers cannot charge more than a service package over a working day. Constraints (61) signify that each caregiver has to work for a restricted number of days. Constraints (62) and (63) restrict the caregivers' daily working hours. The daily traveling distance of each caregiver is calculated by Constraints (64). Constraints (65) imply the domain of the proposed decision variable.

The outcome of the final step is the initial planning that the integrated MIP model presented in Section 3.1.1 aims to find. To determine

the updated planning using the matheuristic algorithm, some slight changes need to be made to the steps mentioned above. These changes are listed in the next section.

4.5. Matheuristic algorithm for the updated planning

Regarding the differences between the initial and the updated planning, the changes listed below must be applied in the models presented in Sections 4.1.1 to 4.1.3 so that the final step obtains the updated planning for the second part of the planning horizon:

1. In all three steps, sets J , D , and S have to be replaced with $\hat{J}$, $\hat{D}$, and $\hat{S}$, respectively.
2. In the third step, the objective functions (66) and (67) substitute for objective (1) and lead to a bi-objective mathematical model for the assignment of the packages. The objective function (66) aims at maximizing the total caregiver-patient compatibility by assigning caregivers to packages with which they have high compatibility. On the other hand, the objective function (67) tries to minimize the deviation of the updated planning from the initial one.
3. A_{igd} for days during the second part of the planning horizon ($d \in \hat{D}$) resulting from the initial planning is used as an input parameter ($\hat{A}_{igd}$) in the third step.
4. The parameters RS_{jpd} , RI_{jp} , and RIN_{jpd} are replaced by $\hat{RS}_{jpd}$, $\hat{RI}_{jp}$, and $\hat{RIN}_{jpd}$, respectively, to devise the updated planning based on the revised table of requests.
5. In the third step, Constraints (69) are used instead of Constraints (61) to ensure that for each caregiver, the summation of the working days in the first and the second part of the planning horizon does not outnumber the maximum permitted working days.
6. Constraints (70) are added to the third step's model to ensure that caregivers are not assigned to any service on their off days.

$$\text{Max} \sum_{i \in I} \sum_{g \in G} \sum_{d \in \hat{D}} CG_{ig} \times A_{igd} \quad (66)$$

$$\text{Min} \sum_{i \in I} \sum_{g \in G} \sum_{d \in \hat{D}} PE_i \times |A_{igd} - \hat{A}_{igd}| \quad (67)$$

Objective function (67) is non-linear because of the absolute term. Therefore, it needs to be written in the form of Objective function (68) accompanied by Constraints (71) and (72). Accordingly, the final formulation for the third step of the matheuristic algorithm that aims to obtain updated planning is as follows:

Objective (66)

$$\text{Min} \sum_{i \in I} \sum_{g \in G} \sum_{d \in \hat{D}} PE_i \times (A_{igd}^+ + A_{igd}^-) \quad (68)$$

s.t.

Constraints (58) to (60), Constraints (62) to (65), Constraints (16)

$$\sum_{g \in G} \sum_{d \in \hat{D}} A_{igd} + \sum_{g \in G} \sum_{d=1}^e A_{igd} \leq MD \quad \forall i \in I \quad (69)$$

$$\sum_{g \in G} A_{igd} \leq M \times (1 - OFF_{id}) \quad \forall i \in I, d \in \hat{D} \quad (70)$$

$$A_{igd} - \hat{A}_{igd} = A_{igd}^+ - A_{igd}^- \quad \forall i \in I, g \in G, d \in \hat{D} \quad (71)$$

$$A_{igd}^+, A_{igd}^- \in \{0, 1\} \quad \forall i \in I, g \in G, d \in \hat{D} \quad (72)$$

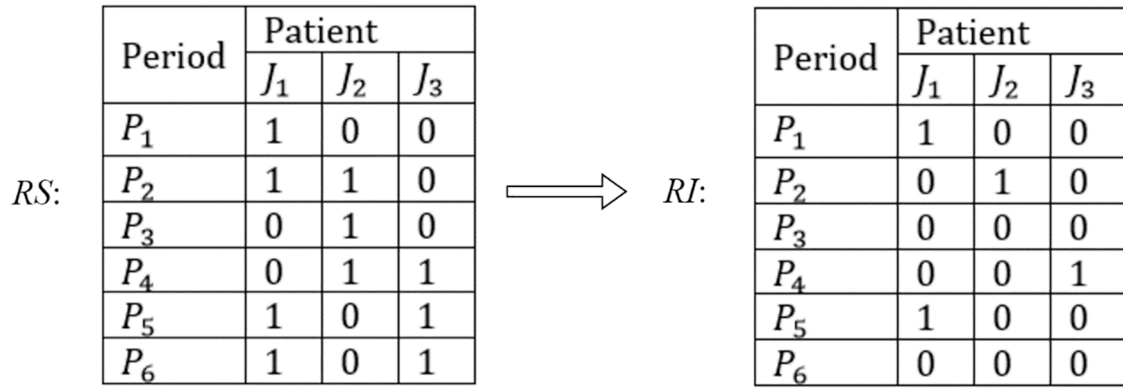


Fig. 2. Extracting matrix RI from matrix RS.

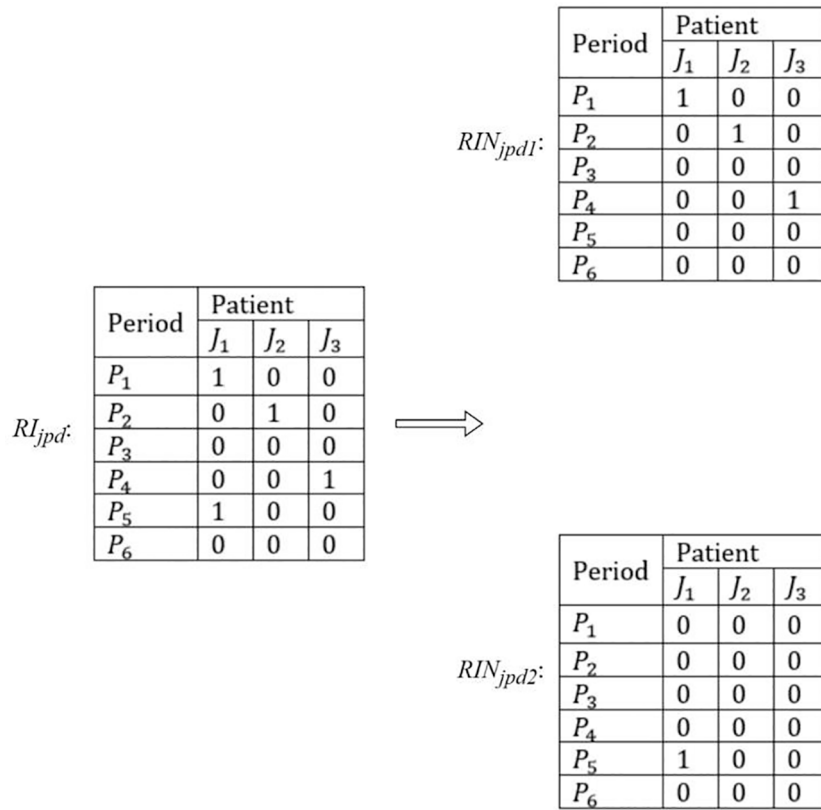


Fig. 3. Extracting matrix RIN from matrix RI.

5. Numerical experiments

To investigate the integrated MIP model's validity and applicability, a case study is carried out in the [Erfan Salamat health center](#). Furthermore, the efficiency of the combinatorial solution algorithm is evaluated using several test problems of different sizes. Both the integrated model and the combinatorial algorithm are coded in GAMS 29.1.0 and solved by the ILOG CPLEX solver on a Laptop with a 3 GHz Intel® 5500u processor and 8 GB of RAM.

5.1. Case study

The [Erfan Salamat health center](#) is located in Tehran, Iran's capital city. With an approximate area of 751 km² and a population of over 8 million, Tehran is known as one of the major demand points for

healthcare services in Iran. This city is divided into 22 districts, represented in [Fig. 4](#).

5.1.1. Input data

The locations of patients' homes are the first set of input data needed to solve the integrated MIP model. This paper uses google maps to measure the distances between the patients' homes, which are available in Appendix B. Also, [Fig. 5](#) demonstrates the distribution of patients who ask for medical services during the seven-day planning horizon.

[Table 4](#) presents another set of input data, which are the requested services in different periods of the planning days. These services must be carried out by the health center's caregivers, including two nurses (I_1 and I_2) and four auxiliary nurses (J_3 to J_6), each of whom is not permitted to work for more than three daily periods ($MP = 3$). In this table, bold services need to be done by a nurse. For example, in the third period (9:00-10:00) of the first day, patients J_1 and J_5 have to be visited by an

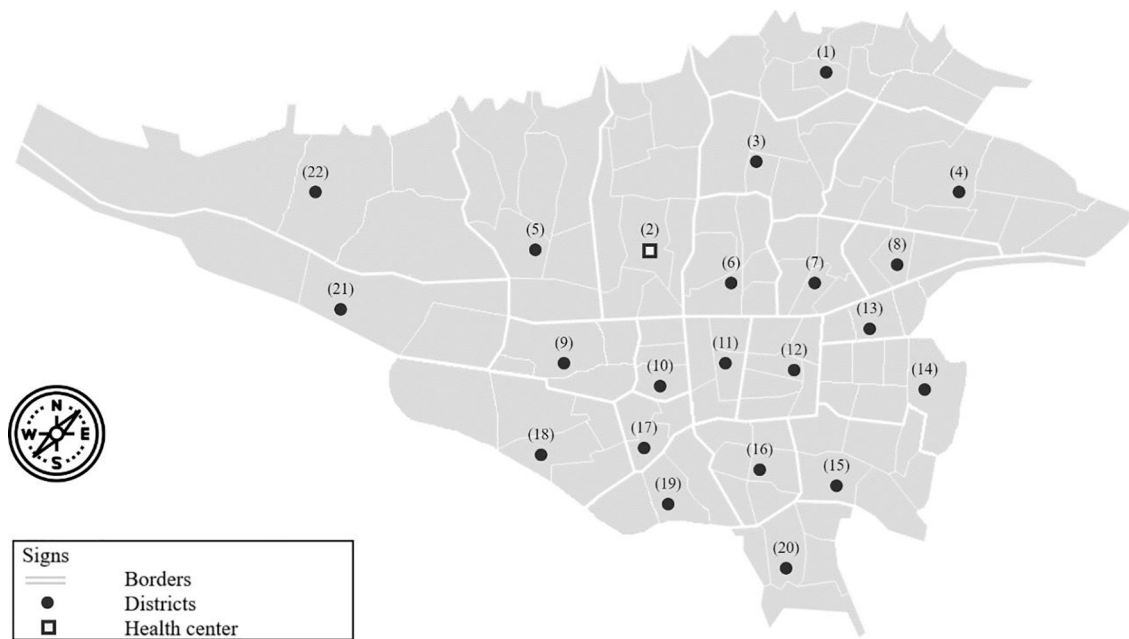


Fig. 4. Geographical map of Tehran province.

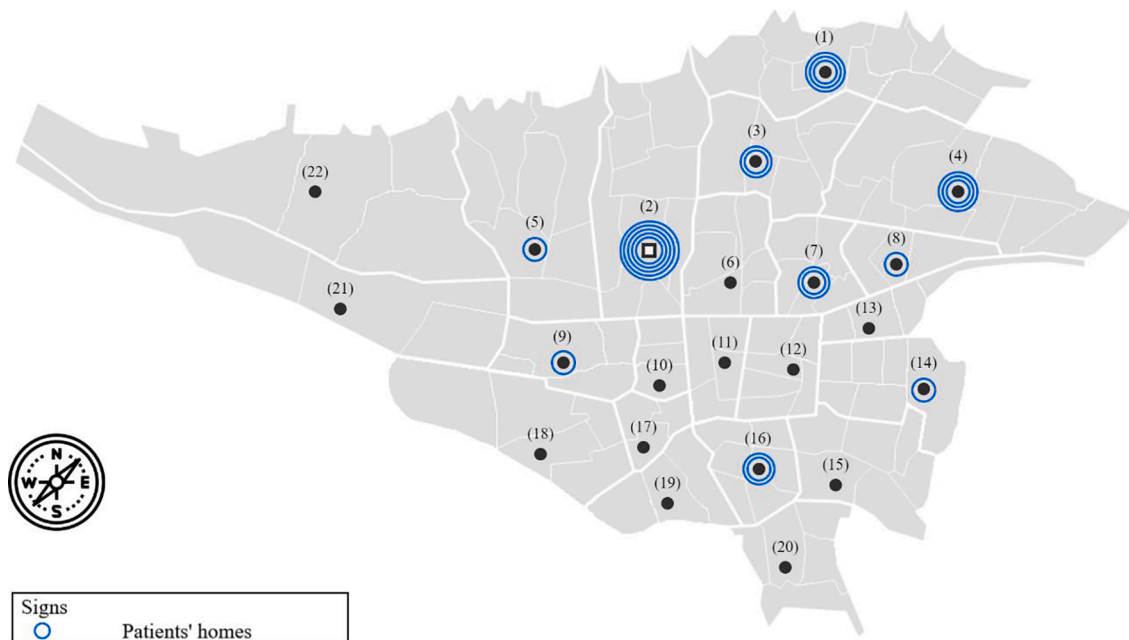


Fig. 5. Patients' distribution over different districts.

auxiliary nurse and a nurse, respectively.

Table 5 is the updated table of requests, which takes the changes registered during the first part of the planning horizon (D_1 and D_2) into consideration. These changes are listed below:

1. Visit cancellations:

- Patient J_{10} cancels his visit in period P_{13} on day D_3 .
- Patient J_{14} cancels his visit in period P_5 on day D_6 .
- Patient J_5 cancels his visit in period P_6 on day D_7 .
- Patient J_4 cancels his visit in period P_{11} on day D_7 .

2. Changes in the pre-determined appointment times:

- Patient J_1 changes his appointment from period P_3 on day D_3 , to the same period on the next day.
- Patient J_{20} changes his appointment from period P_8 on day D_3 , to the previous period on the same day.
- Patient J_6 changes his appointment from period P_{12} on day D_4 , to period P_7 on day D_7 .
- Patient J_3 changes his appointment from period P_4 on day D_6 , to period P_2 on the previous day.
- Patient J_1 changes his appointment from period P_3 on day D_7 , to the next period on the same day.

3. Newly-added requests:

- Patient J_2 requests a new visit in period P_9 on days D_6 and D_7 .

Table 4

Requested services over the planning days.

Period	Day						
	D_1	D_2	D_3	D_4	D_5	D_6	D_7
P_1	$[J_8]$		$[J_8]$			$[J_{11}]$	
P_2		$[J_{18}]$		$[J_{18}]$		$[J_{11}, J_{18}]$	
P_3	$[J_1, J_5]$		$[J_1]$		$[J_1]$		$[J_1]$
P_4	$[J_{12}]$	$[J_{12}]$				$[J_3]$	
P_5	$[J_4]$			$[J_{14}]$	$[J_{14}]$	$[J_{14}]$	
P_6		$[J_{15}]$		$[J_5]$	$[J_7, J_{15}]$		$[J_5]$
P_7		$[J_4, J_7]$			$[J_7]$		
P_8			$[J_{20}]$			$[J_3, J_{20}]$	
P_9		$[J_{11}]$	$[J_{15}]$	$[J_2]$			
P_{10}	$[J_{20}]$			$[J_2, J_{20}]$			$[J_{16}]$
P_{11}			$[J_9]$		$[J_{21}]$	$[J_{21}]$	$[J_4, J_9]$
P_{12}		$[J_6]$		$[J_6]$			$[J_{19}]$
P_{13}	$[J_{10}, J_{16}]$		$[J_{10}]$		$[J_{15}]$	$[J_6]$	
P_{14}	$[J_{17}]$			$[J_{17}]$	$[J_{10}]$	$[J_3]$	$[J_{17}]$

Table 5

Updated patients' requests over the second part of the planning horizon.

Period	Day				
	D_3	D_4	D_5	D_6	D_7
P_1	$[J_8]$			$[J_{11}]$	
P_2		$[J_{18}]$	$[J_3]$	$[J_{11}, J_{18}]$	
P_3		$[J_1]$	$[J_1]$		
P_4					$[J_1]$
P_5		$[J_{14}]$	$[J_{14}]$		
P_6		$[J_5]$	$[J_7, J_{15}]$		
P_7	$[J_{20}]$		$[J_7]$		$[J_6]$
P_8				$[J_3, J_{20}]$	
P_9	$[J_{15}]$	$[J_2]$		$[J_2]$	$[J_2]$
P_{10}		$[J_2, J_{20}]$			$[J_{16}]$
P_{11}	$[J_9]$		$[J_{21}]$	$[J_{21}]$	$[J_9]$
P_{12}	$[J_{22}]$				$[J_{19}]$
P_{13}			$[J_{15}]$	$[J_6]$	
P_{14}		$[J_{17}]$	$[J_{10}]$	$[J_3]$	$[J_{17}]$

- The new patient J_{22} requests a visit in period P_{12} on day D_3 .

Table 6 shows the caregivers' off days and their qualification levels. For example, caregiver I_1 is a nurse who is unavailable on days D_3 and D_6 . Thus, no service must be assigned to caregiver I_1 on these two days.

The compatibility data for the caregiver-patient pairs who meet each other over the first part of the planning horizon consists of random integers between 0 and 10. The compatibility level of five is considered for the other pairs, who do not have the chance to meet in person during this time. Moreover, the penalty coefficient PE_i is uniformly distributed within $[100, 400]$.

5.1.2. Output of the integrated MIP model for the case study

For the case study introduced above, the initial planning with the minimum TPD ($Z_1^* = 149$) is obtained by solving the integrated MIP model presented in Section 3.2. The details of the optimum initial planning are reported in Table 7. For example, according to this table, on the first planning day, caregiver I_1 starts his working day by visiting patient J_5 and then travels to patient J_4 's home to do the next visit.

Fig. 6 depicts the NDP solutions for the updated planning found by solving the bi-objective integrated MIP model presented in Section 3.3

Table 6

Caregivers' off days over the second part of the planning horizon.

Qualification level	Caregiver	Day				
		D_3	D_4	D_5	D_6	D_7
Nurse	I_1					
Nurse	I_2					
Auxiliary nurse	I_3					
Auxiliary nurse	I_4					
Auxiliary nurse	I_5					
Auxiliary nurse	I_6					

Table 7

Optimum initial planning.

Caregiver	Day						
	D_1	D_2	D_3	D_4	D_5	D_6	D_7
I_1	$J_5 \rightarrow J_4$		$J_9 \rightarrow J_{10}$	J_{14}	$J_1 \rightarrow J_{15}$		
I_2	J_{10}	$J_{18} \rightarrow J_7$	J_{20}	$J_2 \rightarrow J_{17}$	J_{21}		
I_3	J_8	J_{15}		J_{18}	$J_{14} \rightarrow J_{15}$	$J_{18} \rightarrow J_{14}$	J_{19}
I_4	$J_1 \rightarrow J_{16}$	$J_4 \rightarrow J_6$	J_1			$J_{20} \rightarrow J_{21} \rightarrow J_6$	$J_{16} \rightarrow J_{17}$
I_5	J_{20}		$J_8 \rightarrow J_{15}$	J_{20}	$J_7 \rightarrow J_{10}$	J_{11}	$J_1 \rightarrow J_9$
I_6	$J_{12} \rightarrow J_{17}$	$J_{12} \rightarrow J_{11}$		$J_5 \rightarrow J_6$		J_3	$J_5 \rightarrow J_4$

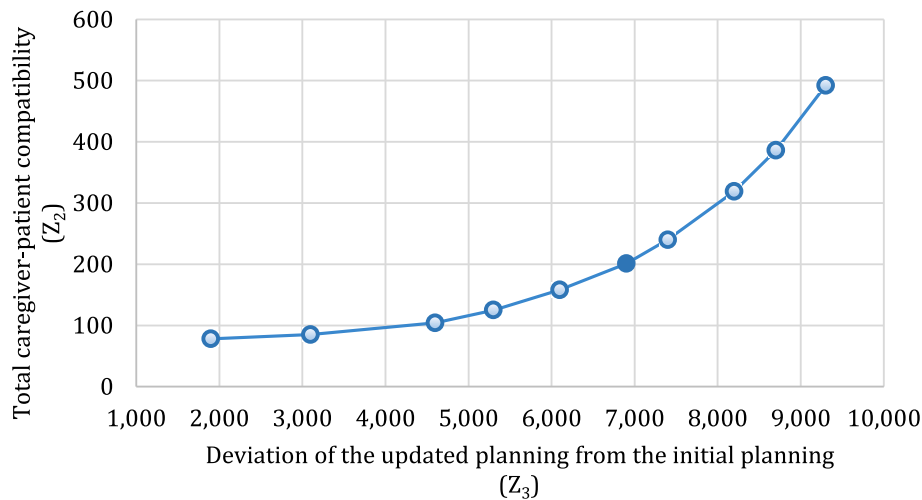


Fig. 6. Pareto-optimal front for the updated planning.

Table 8
Elected updated planning.

Caregiver	Day				
	D_3	D_4	D_5	D_6	D_7
I_1			$J_1 \rightarrow J_{15} \rightarrow J_{21}$		
I_2	$J_{20} \rightarrow J_9 \rightarrow J_{22}$	$J_2 \rightarrow J_{17}$			
I_3			$J_{14} \rightarrow J_{10}$	$J_{18} \rightarrow J_3$	J_{19}
I_4		$J_5 \rightarrow J_{20}$	J_7		$J_6 \rightarrow J_{16} \rightarrow J_{17}$
I_5	$J_8 \rightarrow J_{15}$	$J_1 \rightarrow J_{14}$	$J_3 \rightarrow J_{15}$	$J_{11} \rightarrow J_2$	$J_1 \rightarrow J_2 \rightarrow J_9$
I_6		J_{18}		$J_{20} \rightarrow J_{21} \rightarrow J_6$	

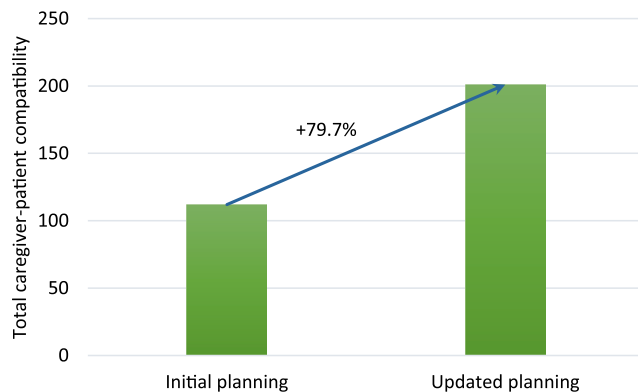


Fig. 7. Improvement in the total caregiver-patient compatibility.

with the AUGMECON2 method.

HHC is almost a new concept in Iran, and not so many people know about it. Thus, the [Erfan Salamat health center's](#) manager believes that to attract new customers, instead of merely caring about making high profits, health centers should pay attention to service quality and satisfaction of caregivers and patients. After sharing the resulting Pareto solutions with the manager, he chose the solution distinguished by the solid blue marker ($Z_2 = 201, Z_3 = 6900$) as a suitable option for the health center since it has a roughly balanced look at the costs and people's satisfaction. Solutions ($Z_2 = 158, Z_3 = 6100$) and ($Z_2 = 240, Z_3 = 7400$) of the Pareto front also seem to be suitable alternatives because of their minor differences with the manager's choice. If a health center only cares about the satisfaction aspect, the rightmost marker is

the most suitable option, while the leftmost marker is the best choice for profit-oriented companies.

Table 8 provides the details of the updated planning indicated by the elected Pareto solution. Regarding this table, all changes in patients' requests and the caregivers' off days are respected in the planning.

Fig. 7 demonstrates the significant superiority of the updated planning over the initial planning in the total compatibility level. This figure reveals that although no compatibility data is available for the first part of the planning horizon, the updated planning has a desirable performance in achieving caregivers' and patients' satisfaction during the second part.

The [Erfan Salamat Health Center](#) prepares the weekly working plans of the caregivers manually and only focuses on minimizing the TPD. The center's suggested planning for the studied week can be found in Appendix B. The center's initial and updated planning have the TPDs of 216 km and 257.5 km, respectively. These values are both significantly greater than those of the integrated MIP model ($Z_1^* = 149$ for the initial planning and $Z_1 = 212.5$ for the elected updated planning), even though minimizing the TPD is not the updated planning's aim.

5.2. Sensitivity analysis

This section aims to analyze how the fluctuations in some structural parameters affect each of the objective function values (OFV) for the case study. Figs. 8–12 reveal the outcomes of this analysis. Based on Fig. 8, the TPD per demand ($\frac{TPD}{|S|}$) follows an upward trend as the number of demands ($|S|$) increases. Because of the limited number of caregivers, the rise in demands reduces the model's flexibility in assigning each caregiver to patients who live in the same or adjacent districts, leading

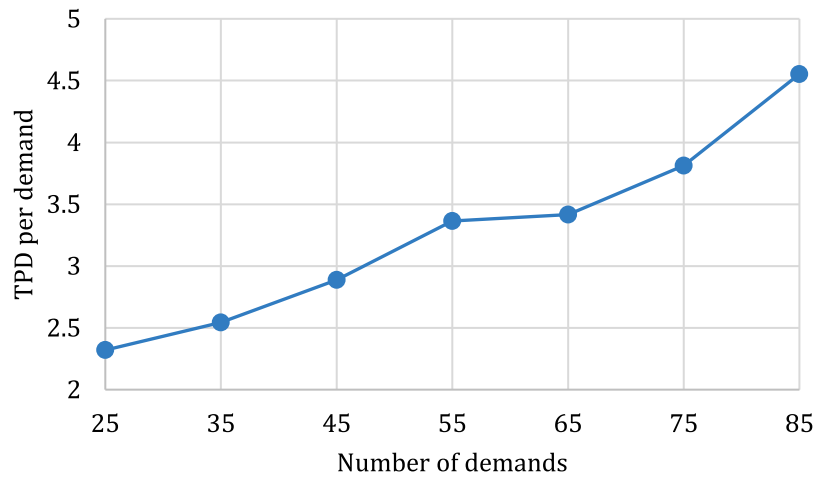


Fig. 8. Changes in the TPD per demand, according to the variations in the number of demands.

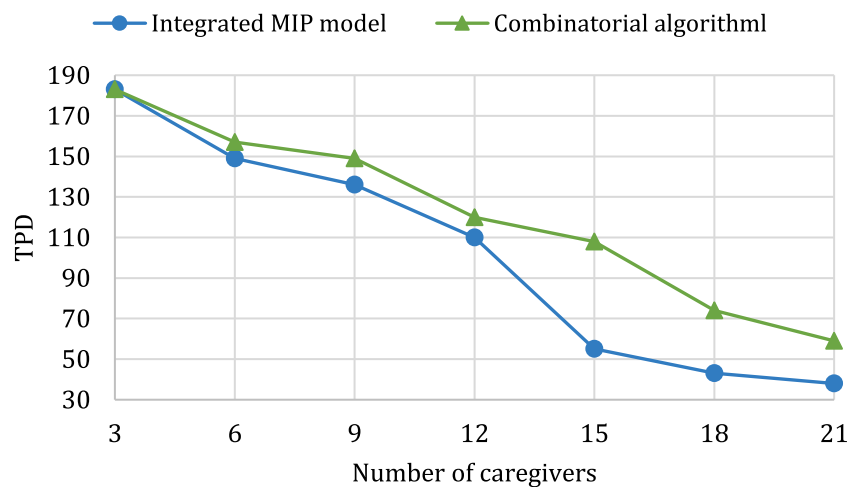


Fig. 9. Changes in the TPD, according to the variations in the number of caregivers.

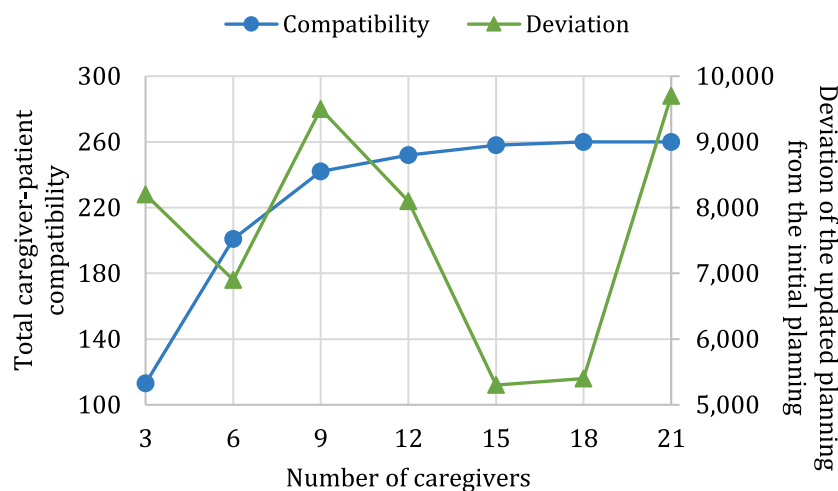


Fig. 10. Changes in the compatibility and deviation, according to the variations in the number of caregivers.

to higher TPD values per demand.

In Figs. 9 and 10, the number of caregivers ($|I|$) varies between 3 and 21 with steps of 3, including one nurse and two auxiliary nurses. As Fig. 9 illustrates, the united MIP model and the proposed combinatorial

algorithm have almost the same behavior regarding the number of caregivers. The more caregivers are available for both of these approaches, the lower the TPD (Z_1) becomes, which seems utterly rational regarding the argument provided for the previous sensitivity analysis.

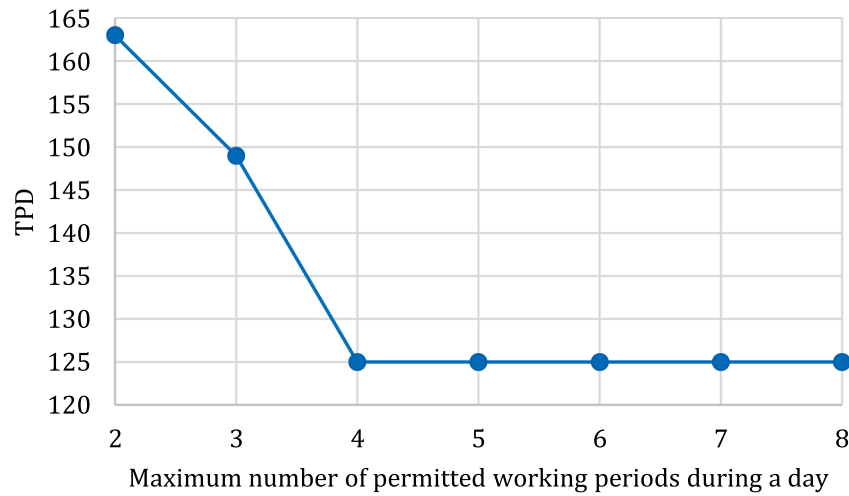


Fig. 11. Changes in the TPD, according to the variations in the maximum number of permitted working periods during a day.

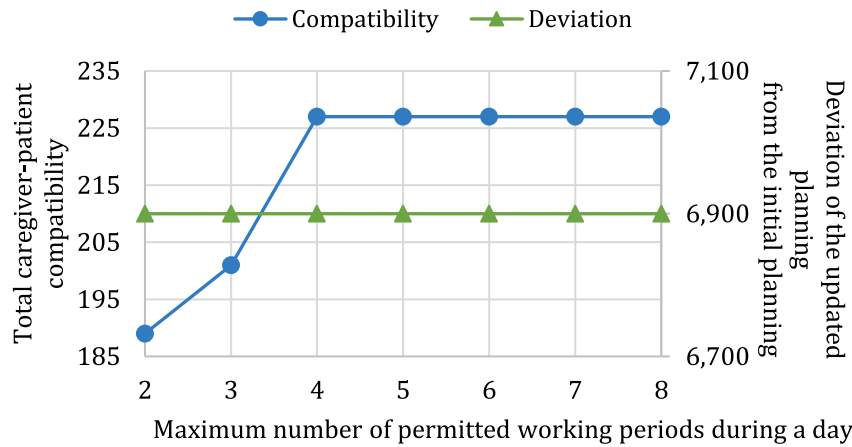


Fig. 12. Changes in the compatibility and deviation, according to the variations in the maximum number of permitted working periods during a day.

Table 9
Test problems.

Problem size	Problem No.	Dimension	
		$ S $	$ I $
Small	1	10	3
	2	15	3
	3	20	3
	4	25	6
Medium	5	40	6
	6	50	6
	7	60	9
	8	70	9
Large	9	90	9
	10	110	12
	11	130	12
	12	150	12

However, it can be seen that the united MIP is superior to the combinatorial algorithm in terms of minimizing TDP since minimizing TDP does not have the highest priority in the latter method. According to Fig. 10, the total caregiver-patient compatibility (Z_2) also positively correlates with the number of caregivers; however, the OFV stops improving when the number of caregivers exceeds 15. On the other hand, the deviation of the updated planning from the initial one (Z_3) fluctuates with no specific pattern.

Figs. 11 and 12 show changes in the OFVs when different values are

assumed for the maximum number of permitted working periods during a day (MP). Based on Fig. 11, despite the considerable sensitivity of the TPD to MP values less than 4, the OFV remains unchanged when the parameter exceeds this value. This specific value determines the impact threshold of parameter MP on the TPD. As Fig. 12 illustrates, MP affects the total caregiver-patient compatibility similarly, while it has no influence over the deviation of the updated planning from the initial one. This insensitivity is that the deviation objective function is entirely dependent on the initial planning. Hence, changes in MP do not affect the OFV unless they are made to the initial planning.

5.3. Efficiency of the proposed solution algorithm

This section uses several test problems of different sizes to evaluate the combinatorial solution method's efficiency.

5.3.1. Test problems

Table 9 shows 12 test problems of small, medium, and large sizes inspired by the real-world benchmark of Bredström and Rönnqvist (2008). These test problems are different in terms of the number of demands ($|S|$) and the number of caregivers ($|I|$). The parameter DI_{jk} follows the uniform distribution, where all values are between one and 32. Moreover, PS_{jds} takes integer values from one to three with equal probabilities. The rest of the parameters are the same as the case study mentioned above.

Table 10

Results of determining the initial planning for the test problems.

NO.	Dimension	Integrated MIP model		Combinatorial algorithm		GAP (%)	Time ratio
		OFV	Time (s)	OFV	Time (s)		
1	(10, 3)	27	3	27	1	0	0.33
2	(15, 3)	32	38	32	7	0	0.18
3	(20, 3)	46.5	143	46.5	22	0	0.15
4	(25, 6)	65	312	65	63	0	0.20
5	(40, 6)	101	529	101	87	0	0.16
6	(50, 6)	119	990	124	184	4.20	0.18
7	(60, 9)	140	1683	147	345	5.00	0.20
8	(70, 9)	166	3318	171.5	521	3.31	0.15
9	(90, 9)	189	6324	196	1197	3.70	0.19
10	(110, 12)	222	10475	233	1900	4.95	0.18
11	(130, 12)	257.5	15441	268.5	2605	4.27	0.17
12	(150, 12)	284	21392	295.5	3558	4.04	0.16
Average			5054		874.16	2.45	0.18

5.3.2. Comparison metrics

To assess the efficiency of the combinatorial solution algorithm, several comparison metrics are used to compare its performance with the integrated MIP model that obtains optimal answers. It should be noted that each solution method has two groups of outcomes, including the initial and updated planning, which need to be compared separately.

CPU time is one of the comparison metrics that refers to the run time of a solution method. The GAP determines the algorithm's error in obtaining the optimal initial planning. This metric is calculated using Equation (73), where LB denotes the lower bound of the objective function and Obj is the OFV of the proposed solution method.

$$GAP = \frac{Obj - LB}{LB} \times 100 \quad (73)$$

Many criteria have been introduced to evaluate the performance of multi-objective algorithms (Zitzler, Deb, and Thiele, 2000). In this study, Quantity Metric (QM), Divergence Metric (DM), and Convergence Metric (CM) are applied to assess the performance of the proposed algorithm in terms of finding the updated planning.

Regarding Equation (74), QM accounts for the ratio between the numbers of NDP solutions achieved by an algorithm and an exact methodology. Thus, algorithms with higher values of QM can be more converged towards the Pareto-optimal front and exact algorithms have a QM of 1.

$$QM = \frac{|\bar{F}|}{|\bar{F}_{exact}|} \quad (74)$$

DM reports the spread of the NDP solutions over the Pareto front and is measured by Equation (75). The higher value of this metric represents a better performance as it implies the algorithm can cover a larger space of solutions.

$$DM = \sqrt{\sum_{k=1}^K \left(\max_{i=1}^N \{f_i^k\} - \min_{i=1}^N \{f_i^k\} \right)^2} \quad (75)$$

where K signifies the number of the objectives, N is the number of NDP solutions, and f_i^k indicates the k -th OFV for Pareto solution i . CM is used to determine the closeness of the solutions to the Pareto-optimal front. This metric is calculated by Equations (76) and (77), where d_i is the minimum distance of NDP solution i from other solutions. For exact algorithms, this metric always equals 0, and for other algorithms, low values of CM denote a relatively good performance.

$$\forall i, j \in \{1, \dots, N\} d_i = \min_{j \neq i} \sum_{k=1}^K |f_j^k - f_i^k| \quad (76)$$

$$CM = \frac{\sqrt{\sum_{i=1}^N d_i^2}}{N} \quad (77)$$

5.3.3. Computational results

The computational results of comparing the solution algorithms by solving the test problems are reported in Tables 10 and 11. A vector $(|S|, |I|)$ is used to demonstrate the dimension of each test problem. In Table 10, the comparison is carried out based on the value of TPD and the computational time. In contrast, in Table 11, QM, DM, and CM substitute for the OFV since instead of a singular answer, a set of Pareto solutions are suggested for the updated planning.

According to Table 10, the combinatorial algorithm obtains the optimal initial planning for the first five test problems. The rest of the problems are solved with a minor error, in which all GAPs are lower than 5%. Moreover, the computational time of the integrated MIP model is

Table 11

Results of determining the updated planning for the test problems.

NO.	Dimension	Integrated MIP model		Combinatorial algorithm				Time ratio
		DM	Time (s)	QM	DM	CM	Time (s)	
1	(10, 3)	6699.60	15	1	6699.60	0	6	0.40
2	(15, 3)	6621.83	212	1	6621.83	0	17	0.08
3	(20, 3)	6754.55	477	1	6754.55	0	54	0.11
4	(25, 6)	6966.04	1645	1	6966.04	0	81	0.04
5	(40, 6)	7259.20	3703	0.70	7052.16	284.52	193	0.05
6	(50, 6)	8037.19	7527	0.85	7411.57	252.14	315	0.04
7	(60, 9)	9112.06	16311	0.80	8375.18	199.33	879	0.05
8	(70, 9)	9442.47	23465	0.90	8291.39	312.65	2550	0.10
9	(90, 9)	9980.65	36500	0.75	8740.02	218.73	4362	0.12
10	(110, 12)	-	> 12 h	-	-	-	7290	< 0.16
11	(130, 12)	-	> 12 h	-	-	-	9996	< 0.23
12	(150, 12)	-	> 12 h	-	-	-	13704	< 0.31
Average		7874.84	> 5 h	0.89	7434.70	140.82	0.91 h	< 0.14

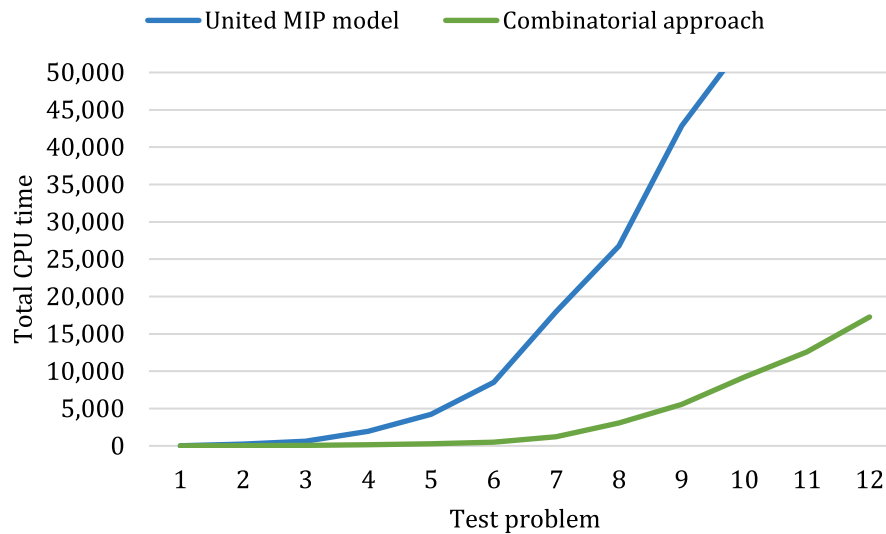


Fig. 13. Comparing the integrated MIP model and the combinatorial algorithm, according to the CPU time.

considerably longer than the other algorithm. For example, the integrated model takes about six hours to solve the 12th test problem; however, the other algorithm solves the problem in less than an hour.

Regarding Table 11, the combinatorial algorithm optimally determines the updated planning for the first four test problems since QM and CM have the values of one and zero, respectively, and DM equals the corresponding values of this metric in the integrated MIP model. Furthermore, the exact methodology needs more than 12 hours to solve the problems larger than (90, 9), while the combinatorial algorithm solves such problems in one to four hours.

Concerning the above-mentioned discussions, it can be concluded that the combinatorial solution algorithm has a reliable and efficient performance in devising the initial and updated planning.

Fig. 13 depicts the superiority of the combinatorial algorithm in the total computational time (the sum of the CPU times in Tables 10 and 11). This superiority gets bolder as the problem size increases because the rate of growth in the computational time is notably lower for the combinatorial algorithm compared to the integrated MIP model.

6. Managerial insights

The following managerial insights can be concluded from the sensitivity analysis implemented in the case study:

- When it comes to devising the updated planning, focusing on its deviation from the initial planning and the caregiver-patient compatibility increases the TPD, which can, in turn, bring early losses to the healthcare center. However, the rise in client satisfaction and staff loyalty compensates for such costs in the long term and shifts the center to the profitability phase.
- To obtain the most favorable number of caregivers, first, the interval through which the total caregiver-patient compatibility is sensitive to this parameter needs to be found. The best number of caregivers is a number out of the interval with the lowest deviation of the updated planning from the initial one. If there is more than one alternative with such characteristics, the option with the least TPD is the most favorable number of caregivers.
- Healthcare centers are recommended to set the maximum number of permitted working periods during a day equal to the impact

threshold of this parameter on the TPD and caregiver-patient compatibility so that these contradictory objectives improve simultaneously.

7. Conclusion

This research made an effort to develop an integrated MIP model for the assignment of the caregivers to the services requested by patients over a planning horizon of multiple days. This model takes two important aspects of the HHCRSP into account, which are the caregiver-patient compatibility and the occurrence of unexpected events. To evaluate the effects of the crucial parameters on the objectives, a sensitivity analysis was conducted on a real-world case study, and some managerial insights were derived accordingly. Due to the high complexity of the integrated model, a semi-exact combinatorial method was proposed to solve the problem over a short time. The programming methods (integrated MIP model and combinatorial method) were implemented by GAMS employing the ILOG CPLEX solver. The precision and computational time of the algorithms were assessed by solving 12 test problems of different sizes. This assessment implies the decent performance of the combinatorial algorithm regarding both of the studied aspects. Future studies can develop hybrid heuristic and meta-heuristic algorithms to solve large-sized instances within lower computational time. They can also address the uncertainty of critical parameters, such as traveling time and service duration.

CRedit authorship contribution statement

Nastaran Oladzad-Abbasabady: Conceptualization, Investigation, Software, Writing – original draft, Data curation, Visualization. **Reza Tavakkoli-Moghaddam:** Methodology, Supervision, Project administration, Writing – review & editing, Resources, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. . Linearization of the non-linear constraints

The following formulations are applied to address the non-linearity of constraints (8), (10), and (14) in the paper:

The linearized equivalent of constraints (8):

$$y_{ijpd} \leq x_{ijpd} \quad (\text{A1})$$

$$y_{ijpd} \leq 1 - x_{ij(p-1)d} \quad (\text{A2})$$

$$y_{ijpd} \geq x_{ijpd} - x_{ij(p-1)d} \quad (\text{A3})$$

The linearized equivalent of constraints (10):

$$y_{ijpd} \leq M \times (1 - a_{ijpd}) \quad (\text{A4})$$

$$y_{ijpd} \geq 1 - M \times (1 - b_{ijpd}) \quad (\text{A5})$$

$$u_{ijpd} \leq M \times (1 - a_{ijpd}) \quad (\text{A6})$$

$$\sum_{k=1}^m \sum_{q=1}^p y_{ikqd} - u_{ijpd} \leq M \times (1 - b_{ijpd}) \quad (\text{A7})$$

$$\sum_{k=1}^m \sum_{q=1}^p y_{ikqd} - u_{ijpd} \geq M \times (b_{ijpd} - 1) \quad (\text{A8})$$

$$a_{ijpd} + b_{ijpd} = 1 \quad (\text{A9})$$

$$a_{ijpd}, b_{ijpd} \in \{0, 1\} \quad (\text{A.10})$$

The linearized equivalent of constraints (14):

$$\gamma_{ijkds} \leq v_{ijds} \quad (\text{A.11})$$

$$\gamma_{ijkds} \leq 1 - v_{ijds} \quad (\text{A.12})$$

$$\gamma_{ijkds} \geq x_{ijpd} - v_{ijds} \quad (\text{A.13})$$

Appendix B. Rest of the case study tables

Table B1
Distances between patients' homes.

District 1	J_1	J_1																				
	J_2	1	J_2																			
	J_3	1.5	2	J_3																		
District 2	J_4	22	19	20	J_4																	
	J_5	20	22	19	2.5	J_5																
	J_6	21	21	19	1	2	J_6															
	J_7	22	20	22	3	1.5	3	J_7														
	J_8	23	22	21	1	3.5	2	2	J_8													
District 3	J_9	7	5.5	6	10	9	11	10	9	J_9												
	J_{10}	7.5	6	7	14	15	13	13	16	2	J_{10}											
District 4	J_{11}	13	11	10	23	20	23	21	20	14	12	J_{11}										
	J_{12}	10	14	12	20	21	21	22	20	14	14	3.5	J_{12}									
	J_{13}	13	10	12	21	21	24	24	21	15	16	2	1.5	J_{13}								
District 5	J_{14}	19	23	21	8.5	7	7	7.5	6	13	10	24	20	22	J_{14}							
District 7	J_{15}	14	16	14	11	15	14	13	10	8	9	13	12	14	16	J_{15}						
	J_{16}	11	8	9	12	10	12	11	13	8.5	8	14	15	12	17	2.5	J_{16}					
District 8	J_{17}	15	14	11	18	16	17	16	14	13	12	6.5	7.5	6	21	6.5	7	J_{17}				
District 9	J_{18}	30	29	26	13	13	10	12	15	19	20	30	32	30	5	23	21	25	J_{18}			
District 14	J_{19}	23	22	22	26	24	25	23	22	21	22	11	15	13	32	13	10	7.5	33	J_{19}		
District 16	J_{20}	27	29	26	20	19	19	20	21	18	20	23	24	22	25	14	13	20	24	15	J_{20}	
	J_{21}	28	28	29	20	23	18	22	22	19	17	25	28	26	24	12	10	18	22	13	1	J_{21}

Table B2

Initial planning devised by the Erfan Salamat Health Center.

Caregiver	Day						
	D_1	D_2	D_3	D_4	D_5	D_6	D_7
I_1	J_{10}	$J_{18} \rightarrow J_7$	$J_9 \rightarrow J_{10}$	J_{17}	$J_1 \rightarrow J_{15}$		
I_2	$J_5 \rightarrow J_4$		J_{20}	$J_2 \rightarrow J_{14}$	J_{21}		
I_3	$J_1 \rightarrow J_{12}$	J_{11}	J_{15}	J_{20}	$J_{14} \rightarrow J_{15}$	$J_{18} \rightarrow J_6$	
I_4	J_8	$J_4 \rightarrow J_6$		J_{18}		J_3	$J_{16} \rightarrow J_9$
I_5	$J_{20} \rightarrow J_{16}$		$J_1 \rightarrow J_8$		$J_7 \rightarrow J_{10}$	$J_{11} \rightarrow J_{20} \rightarrow J_{21}$	$J_{19} \rightarrow J_{17}$
I_6	J_{17}	$J_{15} \rightarrow J_{12}$		$J_5 \rightarrow J_6$		J_{14}	$J_1 \rightarrow J_5 \rightarrow J_4$

Table B3

Updated planning devised by the Erfan Salamat Health Center.

Caregiver	Day				
	D_3	D_4	D_5	D_6	D_7
I_1			$J_1 \rightarrow J_{15} \rightarrow J_{21}$		
I_2	$J_{20} \rightarrow J_9 \rightarrow J_{22}$	$J_2 \rightarrow J_{17}$			
I_3			$J_3 \rightarrow J_7$	$J_{18} \rightarrow J_3$	$J_1 \rightarrow J_6 \rightarrow J_2$
I_4	J_{15}	$J_1 \rightarrow J_5$	J_{14}		$J_{16} \rightarrow J_9 \rightarrow J_{17}$
I_5		J_{18}	$J_{15} \rightarrow J_{10}$	$J_{20} \rightarrow J_{21}$	J_{19}
I_6	J_8	$J_{14} \rightarrow J_{20}$		$J_{11} \rightarrow J_2 \rightarrow J_6$	

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